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Machine check Architecture.

-> It represents rules/contract between hardware & software (OS/BIOS) with regard logging & Signaling error.

-> CPOID EDX MCE & MCA INT 18

-> Machine check global registers include MCG_CAP, MCG_STATUS & MCG_CTL

Interrupts:-

NMI & SMI

-> MCG_CAP -> It provide information about Machine check Architecture implementation in processor

-> MCG_STATUS -> current state of processor after MCE occured

-> MCG_CTL -> controls the reporting of MCE's means enables machine check features & 0's means disables Machinchek features

Error Signaling:-

CMCI - Correctable machine check Interrupt -> OS visible external

CSMI - Correctable System Management Interrupt via SMM handler

MCE -

MSMI - Machine check System manable Interrupt

SMI - System Management Interrupt via SMM handler

memory controller & IIO stack may cause an SMI to be trigged in respose to error event

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MCE - broadcasted to all cores in coherent domain

LMCE - Local MCE → no bradcast only sepcic specific thread

SMI - Highest priority Interrupt cpu → System Management Mode

SMM & SMI handler in Bios.

Bios process the error & expose(or) modify (&) hide event from OS.

Error types:

UC - Uncorrected Error

SRAR - Software recoverable Action Required

SRAO - Software recoverable Action optional

UCNA - Uncorrected No Action

CE - Corrected Error.

MCI_STATUS register → UC, PCC, S & AR

SRAR/SRAO - MCACOD →

SRAR - Dataload, Instruction Access Poison

0134 0150

SRAO -

patrol scrub/ memory sCrub - 00C0 - 00CF

LLC Eviction - 0x017A

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RIPV / EIPV

Recoverable correctable

Recoverable non-correctable

MC Banks — each mc Bank will have 4 MRs

MSR → MCi_status, CTL, ADDR, MISC → Inside

MCi_CTL2 → outside allocation.

error readable after warm reset

→ bog discussion — Call sanitho sh

→ Bios Related —

Ucode - Pcode - Ocode

X86 boot flow

Scandact, ITH, Nano scope.

Bios & FW :

Ras RAS Handling, BIOS Bugs.

→ we inject corrected error. It is expected by bios to take control & clear those errors.

→ MCA: CSMI, CMCI

FW = CSMI, OS - CMCI

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Error flow during RAS

-> Detect - Detect - error sat, global
-> Error should be logged -> internal to IP - IEH
-> Report it rest of the IPs - Core - hirerc
-> Harvesting - what type of error?
-> uncorrected - crashlog -> crashdump
-> Recovery. -> corrm - cold - BMC (external)
-> If FW mod1 -> Bios will handler the error,
OS mod1 -> OS will handler the error
external agent -> BMC will be handler.
OS re

Memory Recovery:-

-> TPR, Self healing technology repair faulty rows maping the failing row & Replacing spare row in the dimm.
-> ECC - Single bit & double bit error:
-> Poison - Uncorrected error - instaded hange will add poison bit & send.
-> Propagater - recive data & sent as UCNA. poision'd drop & resend it.. -> dDR/4
-> command is corrupted viral is reported.
-> Scrubing -> ECC ~~techni~~ techiave peroidraly scans dram & detect & fix soft errors caused by radiations & dat preventig data correption.

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How Fw replay role in RAS

ucode for lot of configuration → for core
pcode for mail box communication → for Punit
ocode for - DOB msm.

→ Booting flow → Ss it starts. → so domain.
x86 boot flow

→ ITh - Intel trace hub. → visA signals.
→ status scope.
→ scan dump.
→ Nano scope.

Boot Process -
1) electricity flow to motherboard & Processor & basic components
2) Processors run basic instruction to BIOS (UEFI)
3.) Power-On self test. → BIOS checks CPU, GPU memory
4.) BIOS looks for device to stage OS.
5.) It load bootloader → for running OS
6.) loads @kernal to memory.

3)

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Boot flow involves

→ Hardware initialization, fW execution, memory setup, device enumaration & OS handoff